Venkata Lakshmi Parimala Pasupuleti

Senior Full-Stack Software Engineer · AI Products · Python · TypeScript · React · LLM & Agentic Workflows · PostgreSQL · AWS · Claude Code & Cursor Daily

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Virginia, USA · open to SF / NYC / remote

PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer building LLM-powered products end-to-end: React frontends, Python/FastAPI backends, PostgreSQL, AWS, and agentic workflows wired to real LLMs in production. LeaseI is a production 8-agent LangGraph system with tool-use orchestration, RAG, durable state, and HITL escalation — agent-based AI in a real product, not a side project (93.6% precision across 1,240 test cases). AgentProbe is a homegrown LLM evaluation framework covering correctness, hallucination rate, latency, and CI regression — exactly what this JD lists as a "nice to have." 8+ years of end-to-end feature ownership in production systems (PostgreSQL 8yr, AWS, Docker, CI/CD). Daily user of Claude Code and Cursor. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and AI-integrated data pipelines; designing agentic components that interact with backend APIs; debugging production reliability issues; using AI coding agents (Claude Code, Cursor) as core workflow tools throughout.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Owned full agentic AI product end-to-end (LangGraph): orchestration, tool use, RAG over regulatory corpus, guardrails as hard enforcement logic, and conditional routing — shipped from spec to production without a manager telling me what to do.

George Mason University
Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries ($R^2 = 0.977$); HuggingFace fine-tuning + PyTorch inference API deployment; statistical modeling and validation. Python + SQL workshops to 600+ students.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Full-stack feature ownership end-to-end** (Python, PHP, Java, Kotlin, JavaScript/TypeScript, Angular, React, PostgreSQL, RabbitMQ, AWS, Kubernetes) — owned 20+ production modules serving 100K+ monthly active users at 99.5% uptime across 120+ Agile sprints. Talked to customers, shaped the spec, shipped to production, iterated on real usage. Operated as de-facto product owner for compliance and financial modules with no PM. Zero compliance violations, zero security incidents across 7 years.
- **Data modeling, APIs, and production operations** — PostgreSQL 8yr: 200+ query optimizations (30% performance gain via execution plan analysis, indexing, partitioning); REST API design across multi-tenant systems; schema migrations across hundreds of thousands of records with rollback preparation; AWS (S3, EC2, RDS, Lambda, Step Functions); Docker + Kubernetes; CI/CD (GitHub Actions) with automated testing; OpenTelemetry + Prometheus + Grafana observability (40% debugging time reduction); Python/Bash automation eliminating 80% of manual workload.

PROJECTS

AgentProbe: LLM Evaluation Framework - Correctness, Hallucination, CI Regression
`LLM-AS-JUDGE` `CI REGRESSION`

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- Homegrown LLM evaluation framework: 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate); LLM-as-judge classifiers; **CI regression harness with golden datasets** so agent behavior can't silently degrade; multi-provider LLM routing (OpenAI/Anthropic/Groq/Ollama) with cost/latency controls; trace-level observability; offline + online evaluation loops. 15-endpoint FastAPI backend. Used to evaluate LeaseI across 1,240+ test cases.

LeaseI: Full-Stack AI Product - React / FastAPI / LangGraph / PostgreSQL-compatible
`8-AGENT LANGGRAPH` `RAG + TOOL USE`

Python · LangGraph · FastAPI · ChromaDB · MCP · SQLite · OpenAI API · LangSmith | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production LLM-powered product built end-to-end: FastAPI backend, 8-agent LangGraph orchestration (tool-use, durable state, RAG over hybrid search corpus, citation grounding, HITL escalation), REST APIs consumed by frontend clients. Designed data models, built evaluation pipeline, shipped to production — 93.6% precision and 87.9% F1 across 1,240 compliance test cases. Evaluated and monitored with AgentProbe.

AI on FHIR: Full-Stack NLP Product - Flask / React / REST APIs
`BERT + EMBEDDINGS` `HIPAA`

Python · Flask · LLMs · BERT · Embeddings · React · REST APIs · Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Full-stack NLP web application: React frontend over Flask backend with REST APIs, LLMs + BERT + embeddings translating natural-language queries into FHIR API calls across structured medical records; HIPAA-compliant RBAC; ~60% reduction in manual retrieval time. Owned end-to-end: data modeling, backend APIs, frontend, deployment.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages & Frontend: Python (8+ yrs), TypeScript, JavaScript, Java (Spring Boot), Kotlin, SQL, Bash — React.js, Redux, Angular, Node.js, FastAPI, Django, GraphQL, REST APIs, OpenAPI

LLM & Agentic: LangGraph, LangChain, LangSmith, MCP, Agentic Workflows, Tool Calling, Orchestration, Durable State, HITL Escalation, RAG Pipelines, Hybrid Search, ChromaDB, Qdrant, Embeddings, Semantic Search, LLM Evaluation, LLM-as-Judge, Hallucination Detection, CI Regression Harnesses, OpenAI API, Anthropic API, Groq, Ollama, Prompt Engineering, Guardrails, Prompt Injection Defense, AI Coding Agents (Claude Code, Cursor)

Backend & Infrastructure: PostgreSQL (8+ yrs, query optimization, schema migrations), MySQL, Redis, RabbitMQ, Apache Kafka, Docker, Kubernetes (K8s), Terraform, AWS (S3, EC2, RDS, Lambda, Bedrock, SageMaker), GCP, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, Distributed Systems, Microservices, Multi-Tenant Architecture, PHI Compliance, HIPAA, SOC 2